Quantification: 699_HYP_HP_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	12
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 12)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1786.67	1403	986.947	px^2
major_axis_length_px	lacuna shape	86.5946	67.1123	47.8852	px
minor_axis_length_px	lacuna shape	35.8136	30.951	8.2955	px
aspect_ratio	lacuna shape	2.4518	1.7046	1.261	unitless
eccentricity	lacuna shape	0.8285	0.8098	0.1393	unitless
solidity	lacuna shape	0.6906	0.6916	0.102	unitless
roots_count	headline	10.1667	11.5	4.745	count
ring_length_r30_px	headline	412.083	381.5	160.869	px
ring_length_r60_px	reported	769.417	731	265.212	px
owned_length_px	ownership-dependent	1172.21	759.263	1118.52	px
edge_count	ownership-dependent	65.25	41.5	64.5659	count
mean_edge_length_px	ownership-dependent	18.5822	17.4601	2.6219	px
perimeter_px	normalised	273.27	188.844	162.22	px
ring_area_r30_px2	normalised	8761.92	7404.5	2899.48	px^2
ring_area_r60_px2	normalised	21065.4	19698.5	4787.84	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0473257	0.0491965	0.0110104	px^-1
ring_density_r60	normalised	0.0368605	0.0366933	0.00915959	px^-1
roots_per_100px_perimeter	normalised	3.9748	3.955	1.6011	per 100 px
ring_attached_length_r30_px	network v2	368.417	348	168.174	px
ring_attached_length_r60_px	network v2	630.583	596	295.44	px
ring_length_w_r30_px	network v2	465.331	433.519	180.832	px
ring_length_w_r60_px	network v2	864.221	824.896	294.971	px
sholl_crossings_r10	network v2	9.3333	8.5	4.2498	count
sholl_crossings_r20	network v2	10.75	11	3.6463	count
sholl_crossings_r30	network v2	10.5833	10.5	3.3155	count
loop_count	loops, report only	4.1667	1.5	6.6583	count
loop_edge_share	loops, report only	0.1603	0.1385	0.1563	share
ring_width_mean_r30_px	width	7.8768	7.8202	0.739	px

Per field
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measure	value	units
analysed_area_px2	869776	px^2
total_skeleton_length_px	19709	px
canalicular_length_density_per_px	0.0226599	px^-1
junction_count	426	count
junction_density_per_px2	0.00048978	px^-2
lacuna_count	12	count

lacunae_per_px2	1.38e-05	px^-2
skeleton_component_count	292	count
mean_component_length_px	67.4966	px
median_component_length_px	16	px
field_length_density_w_per_px	0.025281	px^-1
field_density_without_flagged_per_px	0.0172151	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	563	px
unattached_component_count	29	count
canalicular_length_density_attached_per_px	0.0226599	px^-1
field_length_density_with_unattached_per_px	0.0233071	px^-1
excluded_area_px2	161365	px^2
analysed_field_area_px2	887211	px^2
loop_count	56	count
loop_edge_share	0.1748	share
edge_count	1287	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4712	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	16667	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	971	48.2689	39.9508	1.2082	0.5612	0.6637	-1.1736	126.652	613.53	False
2	1293	67.1123	29.3143	2.2894	0.8996	0.7085	-0.9693	147.124	738.418	False
3	933	52.1712	30.6068	1.7046	0.8098	0.7272	-0.9318	146.222	325.69	False
4	1403	54.0186	34.896	1.548	0.7633	0.888	-0.1596	368.566	862.24	False
5	3093	92.0714	54.1062	1.7017	0.8091	0.6916	1.2856	410.259	393.304	False
6	3594	189.574	42.1988	4.4924	0.9749	0.4917	1.543	562.808	698.994	False
7	2018	112.423	30.231	3.7188	0.9632	0.6731	1.4432	571.53	456.488	False
8	1963	119.951	30.0679	3.9894	0.9681	0.6552	1.3439	639.513	588.421	False
9	812	43.7603	30.951	1.4139	0.7069	0.7167	-1.4386	655.233	193.442	False
10								290.753	563.455	False
11								291.903	675.169	False
12								698.601	352.387	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	12	377	731	915.181	53	17.2676	8.4862
2	12	386	693	717.156	30	23.9052	8.1517
3	8	302	657	1005.88	59	17.0488	9.3181
4	2	209	470	315.622	15	21.0415	6.8106
5	18	813	1542	4371.17	252	17.3459	7.4887

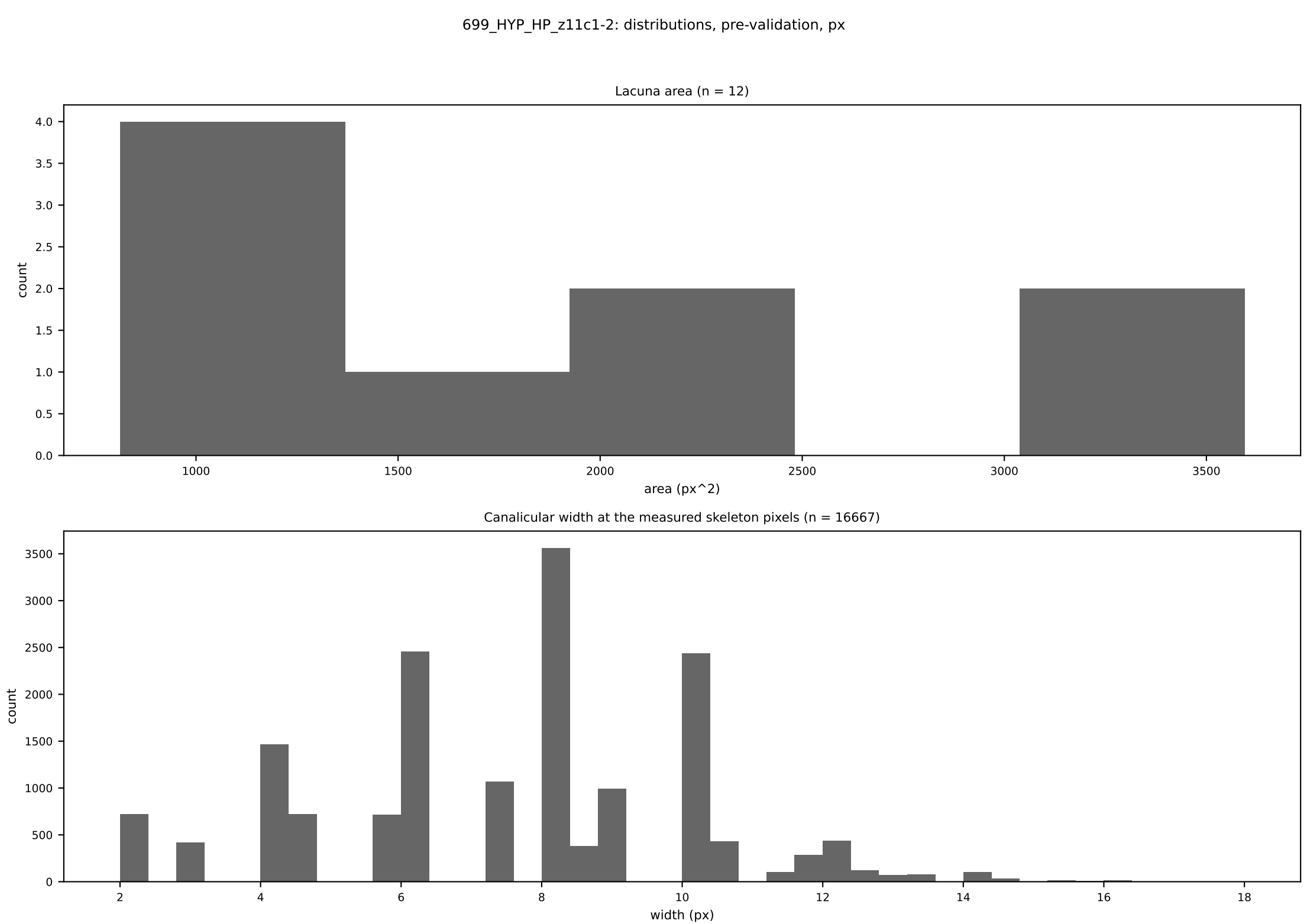
6	14	504	755	554.961	34	16.3224	7.1402
7	13	424	731	728.813	35	20.8232	7.4846
8	13	561	761	632.671	36	17.5742	7.2322
9	10	360	625	789.713	37	21.3436	8.5944
10	11	404	860	1031.68	65	15.8721	8.2415
11	2	228	586	713.784	46	15.517	7.3283
12	7	377	822	2289.87	121	18.9246	8.2448

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	191.622	7458	18845	1	1	0.0505497	0.0387901	6.2623
2	234.25	8068	19848	1	1	0.0478433	0.0349154	5.1227
3	172.995	7102	19616	1	1	0.0425232	0.0334931	4.6244
4	174.066	7351	20270	1	1	0.0284315	0.023187	1.149
5	467.446	11796	28237	1	1	0.0689217	0.0546092	3.8507
6	648.429	15826	32285	1	1	0.0318463	0.0233855	2.1591
7	409.931	10384	22846	1	1	0.0408321	0.0319969	3.1713
8	347.12	10634	19781	1	1	0.0527553	0.0384713	3.7451
9	182.894	6958	19375	1	1	0.051739	0.0322581	5.4676
10	186.066	7155	17832	1	1	0.056464	0.0482279	5.9119
11	91.9828	5369	14348	1	1	0.042466	0.0408419	2.1743
12	172.439	7042	19502	1	1	0.0535359	0.0421495	4.0594

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	348	593	432.99	829.039	8	8	13
2	374	618	434.049	765.115	9	12	12
3	272	599	349.907	743.428	6	9	7
4	169	284	233.752	529.76	4	4	7
5	799	1489	916.37	1721.76	18	18	17
6	374	454	561.688	842.583	14	14	13
7	400	491	468.605	820.754	11	13	7
8	532	610	642.328	862.252	13	13	8
9	321	490	406.563	701.6	8	11	9
10	325	657	450.463	964.21	8	11	13
11	159	521	261.865	661.215	3	8	8
12	348	761	425.392	928.938	10	8	13


699_HYP_HP_z11c1-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 16667)
count
width (px)